

CARSON SHELLENBERGER

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SUMMARY

Hands-on mechanical engineer with 20+ years in medical devices & robotics, driving innovative designs from concept to production. Adept at building and leading small, nimble teams that deliver.

CORE SKILLS

- Product Development: User needs, Product Requirements, DFX, FMEA, Risk Analysis
- Prototyping & Manufacturing: Machining, Molding, Sheet Metal, MIM, Additive, Laser Cutting, Extrusion
- Test & Automation: Custom test rigs, Allen Bradley/Koyo PLCs, sensors, vision systems
- Tools: SolidWorks, Onshape, Minitab, Microsoft, Python, Ladder Logic, Six Sigma Green Belt (NC State), Jira
- Leadership: Cross-functional team building, mentoring, strategic planning
- Compliance: Regulatory requirements, IEC standards, design control, project management

EXPERIENCE

Gilero/Sanner – Durham, NC

Principal Design Engineer | Jul 2025 – Present

- Tech lead for infusion pump system, driving technical project direction and execution
- Electromechanical design of complex drivetrain utilizing steppers and brushed DC motors
- Test method and fixture development including data analysis (python, excel)
- Managed team to define work packages and ensure timely execution to minimize risk

Noah Medical – San Carlos, CA / Cary, NC

Principal Robotic Instruments Engineer | Jan 2023 – May 2025

- Led cost-reduction initiatives for a robotic bronchoscope system across design, packaging, and supply chain
- Developed performance and reliability improvements for cable/coil-pipe mechanisms
- Built automated test and manufacturing systems (Allen Bradley PLCs, HMIs)
- Mentored engineers and guided compliance with IEC standards for electrosurgical tools

Titan Medical – Chapel Hill, NC

Principal Engineer, Instrumentation | Aug 2020 – Jan 2024

- Managed 6-person team to develop robotic surgical instruments from concept to production
- Collaborated with surgeons and marketing to define product & engineering requirements
- Established & managed internal pilot manufacturing line and machine shop
- Led transition to scalable, cost-efficient production

Teleflex Inc. – Morrisville, NC

Principal Engineer, Surgical R&D | Oct 2015 – Aug 2020

- Designed next-gen Hem-o-Lok clip and applier
- Created test methods and fixtures for design verification and validation
- Developed high-volume automation systems and transferred products to OUS production

Solara Automation – Morrisville, NC

Principal Engineer | Jan 2015 – Oct 2015

- Designed custom automation systems across medical, industrial, and consumer markets
- Worked closely with clients to define and deliver tailored solutions

TransEnterix Inc. – Morrisville, NC

Director of Engineering | Mar 2008 – Nov 2014

- Early core team member for SPIDER and SurgiBot surgical platforms
- Led technical efforts from early concept to commercial launch

NeoGuide Systems – Santa Clara, CA

Senior Mechanical Engineer | Mar 2005 – Mar 2008

- Designed force transmission systems for robotic endoscopy

EDUCATION

MS, Mechanical Engineering – Colorado State University

BS, Mathematics – Colorado State University

PATENTS

- **12064115,12102334,11266408, 11160559: Clip Applier having Stabilizing Member**
- **12023041: Clip Applier**
- **11607227: Surgical Clip and Applier**
- **11534177: Flexible Stabilizing Member for a Clip Applier**
- **11291459: Latching Wire Clip**
- **10925616: Clip Applier with Replaceable Tips**
- **11446103: Instruments for Mechanized Surgical System**
- **8192426: Devices and methods for treatment of Luminal Tissue**
- **WO2013116869 A1: Mechanized multi-instrument surgical system**
- **US20110251599 A1: Deflectable Instrument Shafts**
- **WO2005067668 A3: Devices and methods for treatment of luminal tissue**
- **US20120265214 A1: Locking mechanism for deflectable instrument shafts**
- **US20110230723 A1: Active Instrument Port System for Minimally-Invasive Surgery**
- **WO2009100368 A1: A segmented instrument having braking capabilities**
- **US20130281924 A1: Segmented instrument shaft with anti-rotation features**

PORTFOLIO

For a more detailed look at my work experience follow the link or scan the code

<https://cshellenberger.github.io/portfolio/>

